

Project Milestone 4

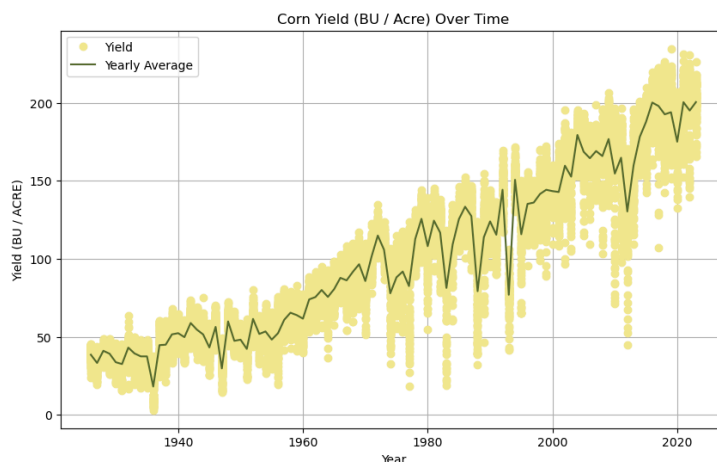
Group 25: Climate/Agriculture

Problem Statement & Research Question

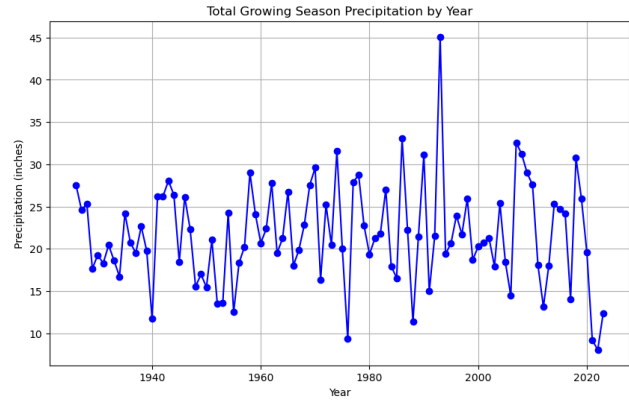
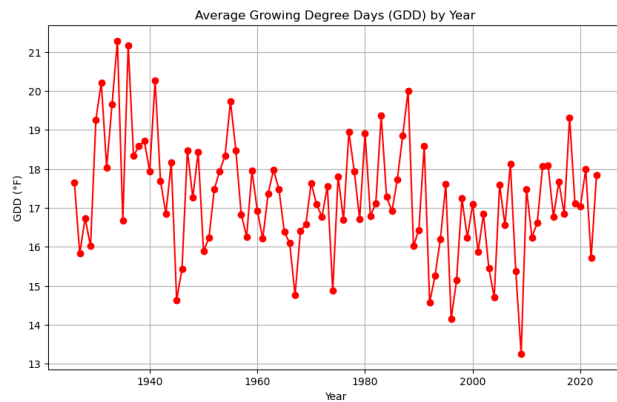
The Earth's climate is changing at unprecedented rates, affecting many regions of the world in dramatic and unpredictable ways. Simultaneously, as the human population grows, the agriculture industry must increase its annual yield to meet the demands of a growing population. However, this becomes more complex as the climate conditions needed to efficiently grow crops on a large scale change in drastic and unexpected ways. Thus, models that seek to understand and quantify the relationship between climate and crop yield become increasingly important to address this challenge and optimize agricultural resource use. In our project, we will be using climate data to predict corn yields in Iowa over the past century as a microcosm for the broader issue of food security in a changing climate.

Research Question: Can climate data be used to predict corn yields in Iowa?

Exploratory Data Analysis (EDA)

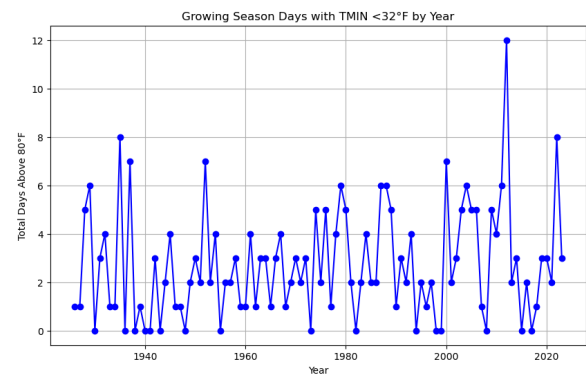
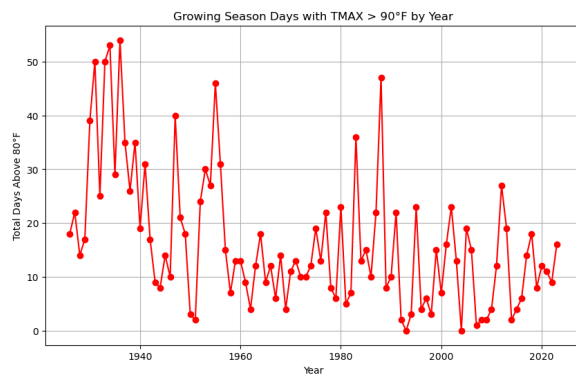


This graph shows the total corn yield (bushels produced / acres planted) in Iowa, plotted as a function of time. It clearly shows a trend of increasing crop yield, likely indicative of improvements in agricultural technology over the past century that have increased the efficiency of each harvest ⁽¹⁾. These advancements have increased the average corn yields from approximately 30 bushels per acre to over 170 bushels per acre in recent years ⁽²⁾. As such, we will be using 'year' as one of our predictors in the model – in addition to all of the climate data – as a proxy for these agrotechnological improvements. We should note that there are also a few pronounced spikes or dips in the average yield that may require further investigation and may be due to extenuating circumstances.



The two graphs above reflect climate data over our given time range, each showing temperature and precipitation trends respectively. The left graph uses average GDD per year, with the GDD for each day found by subtracting the average temperature for the day minus 50. GDD is a common metric used in agriculture when estimating current growth stages of a crop ⁽³⁾. Here, the changes in average yearly GDD simply reflect changes in average temperature throughout the years.

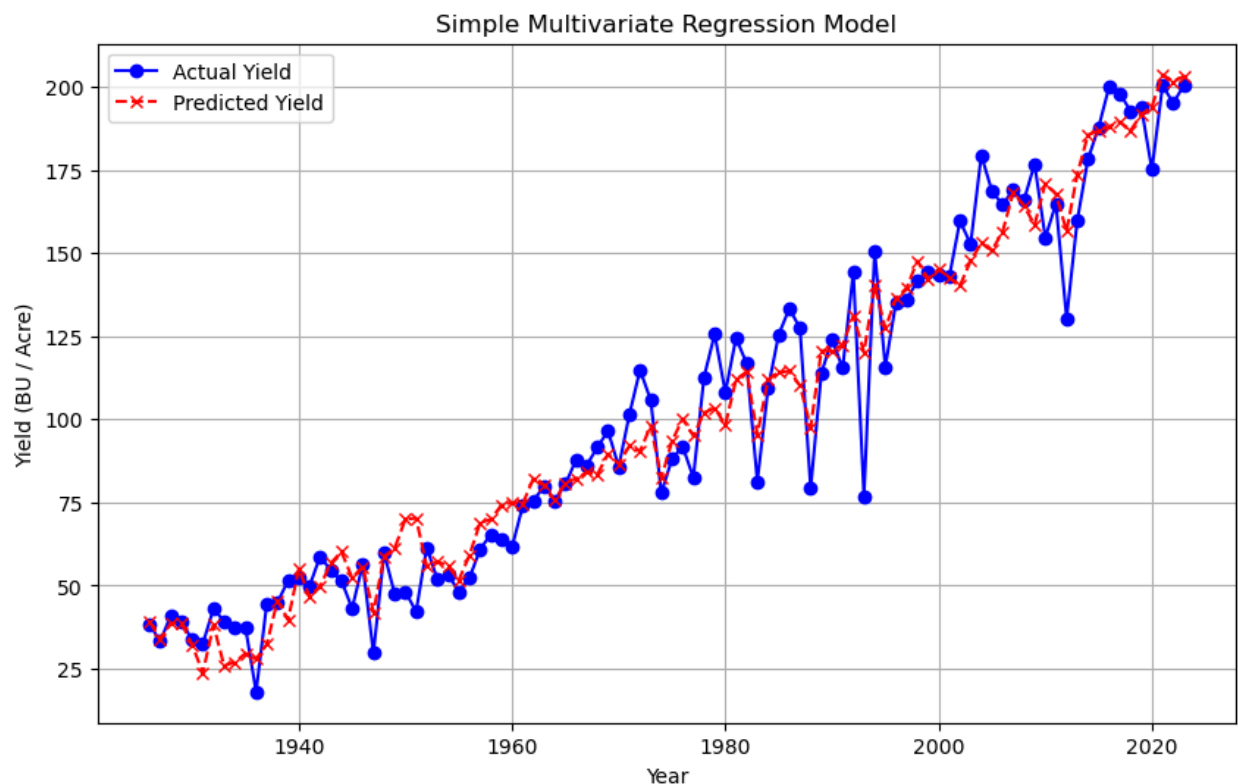
When inspecting the climate data, it is clear that neither temperature nor precipitation exhibit any strong temporal trends over the sampling period. If anything, the growing degree days (GDD) per year are slightly decreasing over time, in line with a slight decrease in average temperature over the same period.⁽⁴⁾ This is further evidence that the increasing yield seen in figure 1 is primarily due to agrotechnology, rather than any climate trends. However, it is likely that much of the ‘noise’ in the yield data is due to the fluctuations in our climate data seen here. The noise in the yield data appears to come from short-term fluctuations in climate data (e.g. heatwaves, droughts, etc). We must note that there may be other factors including increased pest infestations or changes in soil quality that make it difficult to isolate the precise effects of climate on yield outcomes.⁽⁵⁾



Using our conclusions from above, we decided to add new parameters to the data to account for cold shock, heat shock, and precipitation, which can have detrimental effects on the growth of the corn crop. For this, we defined “heat shock days” as days with a maximum temperature $> 90^{\circ}\text{F}$, whereas “cold shock days” were days with a minimum temperature $< 32^{\circ}\text{F}$. Again, there is no obvious temporal trend visible in the data besides a slight decrease in the number of “heat shock days”. Lastly, for our baseline model, we included growing season precipitation as a predictor for our yield as well as an interaction term between our GDD and precipitation predictors.

Baseline Model

For our baseline model, we implemented a multivariate model with the output variable being the crop yield (bushels of corn/acres planted). For the predictor variables, we used forward stepwise selection ($p = 0.1$) to narrow down several predictors. The selected predictors included **Year²**, **Year** (to represent advancements in agricultural technology), **Growing Season Precipitation**, **Hot Shock Days**, **Cold Shock Days**, and an interaction term between **GDD x Precipitation**.



Results

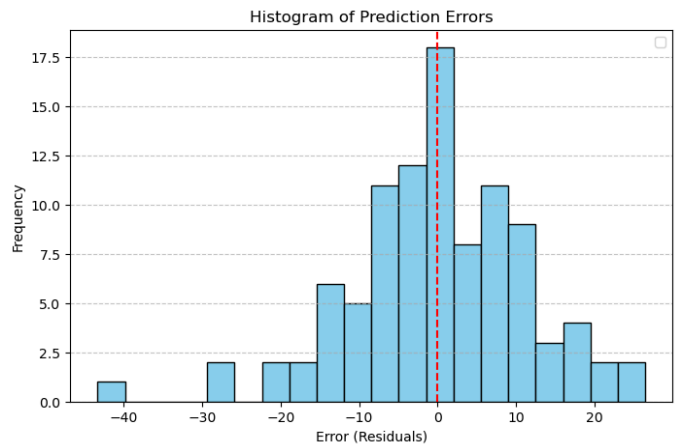
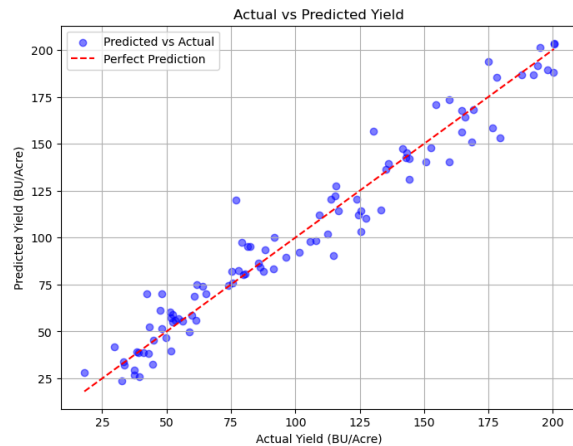
	Coefficient	P-Value
year_sq	33.370864	2.067743e-09
year	13.795778	9.457461e-03
prcp	-11.023236	3.881333e-02
days_above	-9.386725	4.403498e-06
tavg_prcp	8.360655	9.812415e-02
days_below	-1.981968	1.274945e-01

Looking at the coefficients, we arrive at a formula for yield given the normalized variables:

$$\begin{aligned} \text{Yield in Bushels per Acre} = & \\ & 33.37 * \text{Year}^2 \quad + \quad 13.80 * \text{Year} \\ - & 11.02 * \text{Precipitation} \quad + \quad 8.36 * (\text{Growing Degree Days} * \text{Precipitation}) \\ - & 9.39 * \text{Hot Shock Days} \quad - \quad 1.98 * \text{Cold Degree Days} \end{aligned}$$

First, looking at the coefficient for Year^2 , which has the largest coefficient, hinting at its statistical significance. This significance should not be too much of a surprise to us, upon inspection of our year vs yield data (figure 1), we can see the general trend in yield looks to resemble a slight quadratic over time. Moreover, the increasing returns to scale in yield from technological innovation reflect such a significant predictor⁽⁶⁾. The next largest coefficient is that of Year which additionally reflects the importance of technological innovation over time⁽⁷⁾. What is a source of some perplexity when viewed on its own is the coefficient attached to precipitation. The negative value does not make sense as we would expect increased precipitation to yield more crops. However, when taken together with the interaction term of growing degree days and precipitation, a more complete picture begins to emerge. In fact, it is of note that the GDD term on its own was not statistically significant enough to include in our baseline multivariate regression model. Put simply, the interaction term tells us that a GDD is of no use if it is not accompanied by enough precipitation (this, over the course of a whole growing season, of course). Then, we see that on the flip side excess precipitation on its own is detrimental to crop growth, years with droughts sufficiently infrequent so as to bias the coefficient to the positive.

The precipitation and interaction terms both have roughly the same p-value, indicating approximately equal contribution to explaining the variance in yield. Finally, we have the expected signs for both hot and cold shock days, although it is of note that the hot shock day predictor has both a coefficient half an order of magnitude larger and a p-value nearly five orders of magnitude smaller than the cold shock day predictor. All together, these terms constitute the basis of our choice for the final model and associated pipeline.



The graphs above illustrate the actual vs predicted yield for our baseline multivariate model, as well as the overall distribution of errors. It is clear from these graphs that the error is both homoskedastic and normally distributed, validating our primary assumptions when setting up the multivariate regression model.

Final Model Pipeline

We chose to select cross-validated LASSO regression as our final model because it combines both multivariate regression and effective feature selection. Its capability to penalize large coefficients reduces overfitting and helps to identify the most impactful predictors within the dataset. This is particularly advantageous for our climate-based dataset, which includes many potentially correlated predictors as well as interaction terms. Moreover, LASSO's ability to remove unnecessary predictors is particularly applicable for this model, as illustrated above where certain predictors like **GDD** went unselected in our forward stepwise selection.

In order to perform this regression, we have set up the following model pipeline:

1. **Increase data:** Up until this point we have been using climate data from a single weather station centrally located in Iowa and attempted to correlate it with the average overall yield produced by the state. In order to make cross-validation more meaningful (more data within validation sets), we intend to use multiple different weather stations correlated 1:1 with yield values from nearby counties to expand our dataset.
2. **Standardize/Scale Predictors:** Given that the climate variables (e.g., precipitation, growing degree days, temperature extremes) are measured on different scales, it is essential to standardize the predictors to ensure comparability and prevent bias in the regression coefficients. We are also continuing to think about how to best combine the

predictors we have to endogenize the effect of drought (e.g., with a binary indicator variable).

- 3. Perform Lasso Regression with CV and identify best α :** Using LassoCV from Sklearn, we will perform cross validation to identify the best regularization parameter (α) for use in our final Lasso Regression model.
- 4. Create final model using α :** After identifying the best regularization parameter (α), we will use the Lasso package from Sklearn to train our final model.
- 5. Evaluate the Model on the Test Set:** Use a loss function like MSE to evaluate the models performance, and compare with the baseline model above.
- 6. Visualize Results**

Citations

1. Carolyn Dimitri, Anne Effland, and Neilson Conklin. The 20th Century Transformation of U.S. Agriculture and Farm Policy. Economic Information Bulletin No. 3, United States Department of Agriculture, Economic Research Service, June 2005.
2. "Factors Impacted by Increasing Plant Populations in Corn." Crop Science US, Bayer, [https://www.cropscience.bayer.us/articles/bayer/factors-impacted-by-increasing-plant-populations-in-corn](https://www.cropsscience.bayer.us/articles/bayer/factors-impacted-by-increasing-plant-populations-in-corn). Accessed 2 Dec. 2024.
3. Zachary Curtis, Brittany Clark. Penn State Extension. "Understanding Growing Degree Days." Penn State Extension, [https://extension.psu.edu/understanding-growing-degree-days#:~:text=Growing%20degree%20days%20\(GDD\)%2C,reach%20a%20predetermined%20growth%20stage](https://extension.psu.edu/understanding-growing-degree-days#:~:text=Growing%20degree%20days%20(GDD)%2C,reach%20a%20predetermined%20growth%20stage). Accessed 2 Dec. 2024.
4. Lesk, Corey, et al. "Influence of Extreme Weather Disasters on Global Crop Production." *Nature*, vol. 529, 2016, pp. 84–87. <https://www.nature.com/articles/nature16467>
5. Deutsch, Curtis A., et al. "Increase in Crop Losses to Insect Pests in a Warming Climate." *Science*, vol. 361, no. 6405, 2018, pp. 916–919. <https://www.science.org/doi/10.1126/science.aat3466>
6. McKinsey & Company. "Agriculture's Connected Future: How Technology Can Yield New Growth." *McKinsey & Company*, October 2020. <https://www.mckinsey.com/industries/agriculture/our-insights/agricultures-connected-future-how-technology-can-yield-new-growth>
7. U.S. Department of Agriculture Economic Research Service. "Innovations in Seed and Farming Technologies Drive Productivity Gains and Costs on Corn Farms." *USDA ERS*, April 2022. <https://www.ers.usda.gov/amber-waves/2022/april/innovations-in-seed-and-farming-technologies-drive-productivity-gains-and-costs-on-corn-farms/>